



SAPIENZA
UNIVERSITÀ DI ROMA

Ciao Manni ti dedico tutto

Facoltà di Scienze Matematiche, Fisiche e Naturali
Laurea magistrale in Fisica Teorica

Francesco Passante

ID number 2019461

Advisor

Prof. Stefano Giagu

Academic Year 2025/2026

Ciao Manni ti dedico tutto

Master thesis. Sapienza University of Rome

© 2026 Francesco Passante. All rights reserved

This thesis has been typeset by L^AT_EX and the Sapthesis class.

Author's email: francescopassante@icloud.com

Abstract

The aim of the present thesis is to develop a modern machine learning architecture for lattice gauge theories. Gauge equivariant convolutional neural networks have shown that enforcing exact lattice gauge symmetry in a neural network significantly improves regression accuracy on gauge invariant observables. However, their convolutional kernels stay fixed after training and cannot adapt to configuration-dependent features. We introduce GELT (Gauge Equivariant Lattice Transformer), a gauge equivariant attention-based encoder network for lattice gauge theories that exploits the configuration-dependent attention mechanism to focus on the most physically relevant input features.

We test GELT on various regression targets involving physical observables and compare its performance against existing gauge-equivariant architectures. We also investigate the interpretability of the attention filters, studying whether attention localizes in topologically rich regions and whether the attention range correlates with physical correlation length.

Contents

1	Introduction	1
1.1	Motivation: machine learning for lattice gauge theory	1
1.2	From equivariant convolutions to equivariant attention	1
1.3	Contributions	2
1.4	Outline of the thesis	2
2	Lattice gauge theory	3
2.1	The Euclidean path integral	3
2.2	Gauge fields on the lattice	3
2.2.1	Links, plaquettes, and the Wilson action	3
2.2.2	Gauge groups used in this thesis	3
2.2.3	Wilson loops and observables	3
2.3	Masses from Euclidean correlators	4
2.3.1	The transfer matrix and spectral decomposition	4
2.3.2	Zero momentum projection, vacuum subtraction, effective mass	4
2.3.3	Glueballs and why they are hard	4
2.4	Anisotropic lattices	4
3	Monte Carlo methods for lattice fields	5
3.1	Importance sampling and Markov chains	5
3.2	The Metropolis algorithm	5
3.2.1	Validation	5
3.3	Heat-bath and overrelaxation for $SU(2)$	5
3.4	Anisotropic sampling	6
3.5	Error analysis	6
3.5.1	Autocorrelation	6
3.5.2	Jackknife	6
4	Geometric deep learning and gauge-equivariant networks	7
4.1	Symmetry as an inductive bias	7
4.2	Attention and transformers	7
4.3	Gauge-equivariant architectures for the lattice	7
4.3.1	L-CNN	7
4.3.2	Gauge-covariant ResNets and CASK	8
4.4	What convolution cannot give you	8

5	GELT: a gauge-equivariant lattice transformer	9
5.1	Design requirements	9
5.2	Parallel transport on the L^1 -ball	9
5.3	The GEMHSA layer	9
5.3.1	Positional encodings: RoPE vs. offset bias	10
5.4	The full model	10
5.5	Equivariance verification	10
5.6	Computational cost	10
6	Validation and regression experiments	11
6.1	The inductive-bias gap	11
6.2	Experimental protocol	11
6.3	Regression targets	11
6.4	Results: GELT vs. L-CNN vs. CNN	11
6.5	Gauge-invariance in practice	12
7	Glueball spectroscopy with a learned variational operator	13
7.1	The idea: a network as a variational operator	13
7.2	The classical baseline	13
7.2.1	Toolkit	13
7.2.2	Run-by-run: the isotropic dead end and the anisotropic fix	13
7.3	Constraints on the learned operator	14
7.3.1	Making the loss trainable	14
7.4	Training runs	14
7.4.1	Run 4 — thin input: a genuine operator that loses	14
7.4.2	Run 5 — smeared input channels: the physics fix	14
7.5	Quoting the result: cosh fits and the overlap A_0	14
7.6	Replication on an independent ensemble	15
7.7	Open items in this chapter	15
8	Interpreting attention as a physical measurement	17
8.1	Why equivariant attention is interpretable	17
8.2	Study 1: emergent correlation length	17
8.3	Study 2: localization on topological structure	17
8.4	Study 3: head and layer specialization	18
8.5	Validation methodology	18
8.6	Compute constraints	18
9	Conclusions and outlook	19
9.1	Summary of results	19
9.2	Limitations	19
9.3	Future work	19
9.4	Closing perspective	20
A	Notation and lattice conventions	21
A.1	Notation	21
A.2	Lattice conventions	21

B	Equivariance proofs	23
B.1	Gauge covariance of the shortest-path-averaged transport	23
B.2	Equivariance of the GEMHSA layer	23
B.3	Invariance of the readout and the loss	23
C	Codebase and reproducibility	25
C.1	The <code>gelt</code> package	25
C.2	Production runs and hyperparameters	25
C.3	Reproduction recipe	25

Chapter 1

Introduction

1.1 Motivation: machine learning for lattice gauge theory

[TODO: Open with the physics problem: non-perturbative QCD observables (hadron/glueball spectra, topology) are only computable on the lattice, and lattice computations are dominated by two costs – ensemble generation and operator/observable construction. Position ML as a tool for the second: learning better *operators* and *maps* on gauge configurations, not replacing the Monte Carlo.]

[TODO: State the central obstruction: gauge symmetry. A generic network fed raw links must waste capacity learning an exact local symmetry with a group volume growing like $|\mathcal{G}|^V$; a network that is equivariant by construction gets it for free. Preview the “inductive-bias gap” experiment of section 6.1 (plaquette-fed CNN $R^2 \approx 0.99$ vs. link-fed CNN $R^2 \approx 0$ on the same target) as the one-plot motivation.]

1.2 From equivariant convolutions to equivariant attention

[TODO: Brief history: L-CNN [favoni2022lcnn] showed exact lattice gauge equivariance via parallel-transported convolutions and bilinear layers; gauge-covariant ResNets [nagai2021covariant] and the CASK transformer [cask2025] extended the toolbox. Limitation to stress: convolutional kernels are *static* after training – the same stencil everywhere, on every configuration.]

[TODO: The thesis proposal: GELT, a gauge-equivariant attention encoder. Attention weights are configuration-dependent, so the network can allocate its receptive field where the physics is (topological lumps, correlated regions). Two design departures from L-CNN to state up front: (i) the value path is matrix-bilinear ($\alpha Q^\dagger \tilde{V}$), so L-CNN’s loop-growing universality argument transfers; (ii) parallel transport between sites is averaged over *all* shortest lattice paths in the L^1 -ball (a DP recursion,

not enumeration).]

1.3 Contributions

[TODO: Bullet list, one item per delivered result. Current inventory:]

- The GELT architecture (chapter 5): gauge-invariant attention scores, matrix-bilinear values, L^1 -ball shortest-path-averaged transport, RoPE geometric prior; equivariance verified by unit tests and a numerical invariance check.
- A pure-tensor lattice codebase: $\mathbb{Z}_2/\text{SU}(2)$ Metropolis and $\text{SU}(2)$ heat-bath + overrelaxation samplers (with anisotropy), classical glueball toolkit (APE smearing, correlators, GEVP, jackknife), and matched-parameter CNN and L-CNN baselines.
- **The headline physics result** (chapter 7): GELT trained as a variational 0^{++} glueball operator on the Rayleigh loss $-C(1)/C(0)$ saturates the transfer-matrix bound, agrees with the classical GEVP on the mass ($m a_t = 0.332 \pm 0.027$), and *beats* it on ground-state overlap: $A_0 = 0.903 \pm 0.047$ vs. 0.837 ± 0.056 , a correlated $\Delta A_0 = +0.078 \pm 0.022$ (3.6σ) combining the original ensemble and an independent replication.
- [TODO: The interpretability study (chapter 8) – attention maps as measurements – once its results exist.]

1.4 Outline of the thesis

[TODO: One sentence per chapter. Ch. 2 lattice gauge theory; Ch. 3 Monte Carlo methods; Ch. 4 geometric deep learning and prior equivariant architectures; Ch. 5 the GELT architecture; Ch. 6 validation and regression experiments; Ch. 7 glueball spectroscopy; Ch. 8 interpretability; Ch. 9 conclusions and outlook.]

Chapter 2

Lattice gauge theory

2.1 The Euclidean path integral

[TODO: Wick rotation, Euclidean action, path integral as a statistical partition function $Z = \int \mathcal{D}U e^{-S[U]}$; expectation values of observables. This is what licenses Monte Carlo (Ch. 3) and the transfer-matrix reading of correlators (section 2.3).]

2.2 Gauge fields on the lattice

2.2.1 Links, plaquettes, and the Wilson action

[TODO: Define link variables $U_\mu(x) \in G$ as parallel transporters, gauge transformations $U_\mu(x) \rightarrow \Omega(x) U_\mu(x) \Omega^\dagger(x + \hat{\mu})$, the plaquette $P_{\mu\nu}(x) = U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\mu} + \hat{\nu}) U_\nu^\dagger(x)$ and the Wilson action $S = \beta \sum_p \left(1 - \frac{1}{N_c} \text{Re Tr } P\right)$ [wilson1974]. Naive continuum limit. Fix the conventions used by the code (they match the repo's CLAUDE.md): direction-first tensor layout, periodic boundary conditions, $\mu < \nu$ plaquette ordering.]

2.2.2 Gauge groups used in this thesis

[TODO: $\mathbb{Z}_2$ as the debug-friendly testbed (why: exact results, bit-exact float64 invariance tests, cheap), $\text{SU}(2)$ as the first non-abelian production group, the $\text{U}(1)/\text{SU}(3)$ generalisation path. Note the design rule that makes the code group-generic: color axes always present ($n_c = 1$ for $\mathbb{Z}_2$), every product an explicit matrix product, every inverse an explicit dagger.]

2.2.3 Wilson loops and observables

[TODO: Rectangular $R \times T$ Wilson loops, the static potential / confinement reading, and the naive (plaquette-based, clover-free) topological charge density $q(x)$ with $F_{\mu\nu} = (P - P^\dagger)/2i$ – these are the regression targets of chapter 6.]

2.3 Masses from Euclidean correlators

2.3.1 The transfer matrix and spectral decomposition

[TODO: Transfer matrix $\hat{T} = e^{-a_t \hat{H}}$, spectral decomposition of the two-point function $C(\Delta) = \sum_n |\langle 0 | \hat{O} | n \rangle|^2 e^{-m_n \Delta}$ (periodic: cosh). Key inequality for Ch. 7: for ANY operator on a single timeslice, $C(\Delta)/C(0) \leq e^{-m_0 \Delta}$ – the variational bound the Rayleigh loss saturates. Stress that the bound **REQUIRES** the operator to live on one timeslice (this becomes the per-timeslice network constraint of section 7.3).]

2.3.2 Zero momentum projection, vacuum subtraction, effective mass

[TODO: $\bar{O}(t) = \sum_{\vec{x}} O(t, \vec{x})$; the 0^{++} channel has a vacuum expectation value, so the connected correlator needs explicit subtraction. Define $m_{\text{eff}}(\Delta) = \log[C(\Delta)/C(\Delta + 1)]$ and the cosh-corrected version; plateau reading.]

2.3.3 Glueballs and why they are hard

[TODO: What a glueball is; the signal-to-noise problem (signal $\sim e^{-m\Delta}$ against a Δ -independent vacuum noise floor); why heavy masses in lattice units are unresolvable. Sets up smearing, GEVP, and anisotropy as the classical rescue toolkit (section 7.2).]

2.4 Anisotropic lattices

[TODO: Fine temporal spacing $a_t = a_s/\xi$ buys more correlator points per physical decay length – the standard trick for heavy states [morningstar1999]. Tree-level anisotropic Wilson action: $\beta_t = \beta\xi$ on temporal plaquettes, $\beta_s = \beta/\xi$ on spatial ones. Bare vs. renormalized anisotropy (the code measures ξ_R via a Creutz-ratio ratio in `validate_anisotropy.py`); be explicit that this thesis uses the BARE tree-level ξ and does not tune it – a declared limitation (section 9.3).]

Chapter 3

Monte Carlo methods for lattice fields

3.1 Importance sampling and Markov chains

[TODO: Why the path integral cannot be done by uniform sampling; importance sampling w.r.t. $e^{-S[U]}$; Markov chains, detailed balance, ergodicity. Contrast with the Haar-uniform ensemble used as a β -independent sanity check in chapter 6.]

3.2 The Metropolis algorithm

[TODO: Single-site Metropolis with checkerboard vectorisation (why the checkerboard: parallel updates only touch non-interacting sites through the staple). Proposals per group: $\mathbb{Z}_2$ sign flip $U \rightarrow -U$; $SU(2)$ near-identity left multiplication $U \rightarrow VU$. The staple sum as the only action difference needed.]

3.2.1 Validation

[TODO: Report the four-panel sampler validations (figures `sampler_validation_z2.png`, `sampler_validation_su2.png`): thermalisation, 2D β -scan against the EX-ACT mean plaquette ($\tanh \beta$ for $\mathbb{Z}_2$, $I_2(\beta)/I_1(\beta)$ for $SU(2)$), the 3D $\mathbb{Z}_2$ transition near $\beta_c \approx 0.761$ vs. the $SU(2)$ confining crossover, and plaquette autocorrelation.]

3.3 Heat-bath and overrelaxation for $SU(2)$

[TODO: Why Metropolis is not enough for spectroscopy (critical slowing, tuned acceptance). $SU(2)$ heat-bath: staple decomposition $A = kV$, Creutz sampling of the scalar quaternion component [creutz1980], $U' = WV^\dagger$. Overrelaxation $U' = V^\dagger U^\dagger V^\dagger$: microcanonical, action-preserving to machine precision (unit-tested), needs re-projection onto the group. Production recipe: 1 heat-bath + n_{or} OR sweeps. Note the replication-run

lesson: the Creutz rejection loop needs a generous iteration cap (acceptance $\sim \sqrt{\pi/2a}$ collapses for the large temporal staples of an anisotropic lattice – cap raised $100 \rightarrow 1000$ after a production run died on it). SU(2)-only; SU(3) needs Cabibbo–Marinari (section 9.3).]

3.4 Anisotropic sampling

[TODO: How ξ enters the staple (per-plane coupling ratio $\xi^{\pm 1}$ folded into the staple sum, β stays the single overall scale); $\xi = 1$ is bit-exact backward compatible (unit-tested). Report `validate_anisotropy.py`: exact 2D plaquette reproduced at $\xi = 1$, plaquette split $\langle P_{st} \rangle > \langle P_{ss} \rangle$, and the measured renormalized anisotropy ξ_R vs. bare ξ (figure `anisotropy_validation.png`).]

3.5 Error analysis

3.5.1 Autocorrelation

[TODO: Integrated autocorrelation time τ_{int} with Madras–Sokal automatic windowing [madras1988sokal]; the rule $n_{\text{skip}} \gtrsim 2\tau_{\text{int}}$ **OF THE TARGET OBSERVABLE** (the smeared glueball operator, not the plaquette). Report the pre-flight measurement that fixed the production n_{skip} (figure `glueball_autocorrelation.png`).]

3.5.2 Jackknife

[TODO: Leave-one-out and delete-block jackknife; why **BLOCKED** jackknife for correlator data; the correlated-difference jackknife (same configs, take the difference inside each sample) that powers every **GELT-vs-classical** significance statement in chapter 7.]

Chapter 4

Geometric deep learning and gauge-equivariant networks

4.1 Symmetry as an inductive bias

[TODO: The geometric deep learning blueprint [bronstein2021gdl]: symmetry group $\rightarrow$ equivariant layers $\rightarrow$ invariant readout. Definitions of invariance and equivariance ($f(g \cdot x) = g \cdot f(x)$); CNNs as the translation-equivariance special case; why building the symmetry in beats data augmentation for EXACT local symmetries (the gauge group volume grows with the lattice volume – augmentation cannot cover it).]

4.2 Attention and transformers

[TODO: Scaled dot-product attention and multi-head self-attention [vaswani2017attention], kept minimal: queries/keys/values, softmax scores, permutation equivariance, positional encodings (absolute bias vs. rotary/RoPE [su2021rope] – both reappear as GELT variants). The property the thesis exploits: attention weights are CONFIGURATION-DEPENDENT, unlike convolution kernels, and they are readable after the fact – the hook for chapter 8.]

4.3 Gauge-equivariant architectures for the lattice

4.3.1 L-CNN

[TODO: Favoni et al. [favoni2022lcnn] in detail (this is GELT’s scaffold AND its matched-parameter baseline): parallel-transported convolutions (L-Conv), bilinear layers (L-Bilin), trace-gated activations, the loop-doubling universality argument (stacked L-Conv+L-Bilin generates arbitrarily large Wilson loops). The repo implements it faithfully in `gelt/lcnn.py` with axis-aligned transports.]

4.3.2 Gauge-covariant ResNets and CASK

[TODO: Nagai–Tomiya [nagai2021covariant] (covariant smearing-like layers, 4D non-abelian) and CASK [cask2025] (gauge-covariant transformer) – one paragraph each; how GELT differs (invariant SCORES + matrix-bilinear VALUES; all-shortest-path L^1 -ball transport rather than axis-aligned or staple-based).]

4.4 What convolution cannot give you

[TODO: The chapter’s closing argument, setting up Ch. 5: static kernels cannot adapt the receptive field to the configuration; physically interesting structure (topological lumps, critical correlations) is inhomogeneous and β -dependent. Attention is the minimal mechanism that fixes this while keeping exact equivariance – IF scores are gauge invariant and values transform covariantly. That "if" is Ch. 5.]

Chapter 5

GELT: a gauge-equivariant lattice transformer

5.1 Design requirements

[TODO: State the contract up front: (i) exact gauge equivariance of every hidden layer (adjoint transformation $W(x) \rightarrow \Omega(x)W(x)\Omega^\dagger(x)$) and exact invariance of the readout; (ii) configuration-dependent, inspectable attention; (iii) receptive-field growth with loop content sufficient for universality (the L-CNN argument must transfer); (iv) group-generic code ($\mathbb{Z}_2 \rightarrow \text{SU}(N)$ verbatim).]

5.2 Parallel transport on the L^1 -ball

[TODO: The key geometric ingredient. Define the signed offset set $1 \leq |\Delta x|_1 \leq R$; the transport $T_{\Delta x}(x)$ as the **AVERAGE** over all shortest lattice paths (multinomially many), computed by a $|\Delta x|_1$ -ordered dynamic-programming recursion, not path enumeration. Why averaging: restores lattice rotation symmetry that any single canonical path breaks (the `mode="single"` variant exists precisely to A/B this). State and prove gauge covariance $T_{\Delta x}(x) \rightarrow \Omega(x)T_{\Delta x}(x)\Omega^\dagger(x + \Delta x)$; mention the octant identity $T_{-\Delta x}(x) = T_{\Delta x}^\dagger(x - \Delta x)$ as a consistency check (unit-tested), deliberately **NOT** exploited at build time. Memory cost table vs. R and D – the wall that motivates offset-chunked attention (section 9.3).]

5.3 The GEMHSA layer

[TODO: Walk the forward pass (mirrors notes/GELTsummary.md): (1) augment the channel set with daggers and the identity; (2) fused Q, Q_v, K, V projections (channel-mixing only – no color mixing, so covariance is manifest); (3) transport K, V to the query site via $T/T^\dagger$; (4) gauge-INVARIANT score $s = \text{Re Tr}[Q^\dagger \tilde{K}]$ + positional encoding; (5) softmax over offsets; (6) matrix-bilinear value $\sum_{\Delta x} \alpha Q_v^\dagger \tilde{V}$ – the L-Bilin analogue, which doubles

loop length per layer and carries the universality argument; (7) channel mix, residual, trace-gated activation (L-Act). Figure: one block diagram.]

5.3.1 Positional encodings: RoPE vs. offset bias

[TODO: The two variants (`blocks_rope` / `blocks_bias`, $\approx 90\%$ shared code): rotary rotation of the score by Δx vs. a learned per-offset bias. Both act only on the INVARIANT score, so equivariance is untouched – the geometric prior is decoupled from the symmetry. Caveat to resolve before the final write-up: RoPE axis coverage requires $d_{qkv} \geq 2D$ (small d_{qkv} in 4D leaves axes unrotated). [TODO: merge the two variants into one module with a `pos_encoding` switch and parametrize the equivariance tests over both – currently the TRAINED variant (`rope`) is the less-tested one.]]

5.4 The full model

[TODO: `ChannelLift` $\rightarrow$ stacked GEMHSA blocks $\rightarrow$ Trace $\rightarrow$ per-site MLP $\rightarrow$ spatial reduction (mean for global targets, "none" for per-site/per-timeslice outputs – the glueball operator mode). Input: plaquette channels (optionally multiple smearing levels, see section 7.4.2); parameter count vs. the matched L-CNN and CNN baselines (table).]

5.5 Equivariance verification

[TODO: Three tiers, report all: (1) unit tests – GEMHSA end-to-end equivariance, $SU(2)$ complex128 and $\mathbb{Z}_2$ float64 (bit-exact), both gate branches, finite-grad backward on $SU(3)$; (2) full-model invariance check $f(W^\Omega, T^\Omega) \approx f(W, T)$ on random $SU(2)$ gauge transformations; (3) [TODO: the worst-case- Ω stress test in float64 – does not exist yet; run it and quote the drift].]

5.6 Computational cost

[TODO: Scaling in V , R , heads, channels; attention over $|L^1\text{-ball}|$ offsets per site; transport precompute vs. on-the-fly (the glueball training builds 3D transports per batch on the fly). Honest table of wall-clock and memory on the V100 for the production runs.]

Chapter 6

Validation and regression experiments

6.1 The inductive-bias gap

[TODO: The motivating experiment. Setup: Haar-random $\mathbb{Z}_2$ links in 2D, target = Wilson action. Analytic backdrop (worth spelling out): every plaquette is ± 1 with mean zero, plaquette pairs share 0 or 2 links so they are uncorrelated, hence $\text{Var}(S) = L^2$ – so compare via $R^2 = 1 - \text{MSE}/\text{Var}(y)$, never raw MSE across L . Result: a CNN fed PLAQUETTES reaches $R^2 \approx 0.99$ (linear sum of its inputs); the same CNN fed raw LINKS sits at $R^2 \approx 0$. The gap is exactly what the equivariant transport provides. Caveat paragraph: a perfect regressor on Haar data memorises the action FUNCTION; physics (β -dependence) requires the Metropolis data path.]

6.2 Experimental protocol

[TODO: Common recipe for all regression runs: ensembles (Haar vs. Metropolis at fixed β), train/val/test splits, target standardisation, matched parameter counts, optimizer/scheduler, seeds. One table.]

6.3 Regression targets

[TODO: The three physical targets and why each probes something different: Wilson action (linear in plaquettes – sanity floor), rectangular Wilson loops $R \times T$ (requires transport beyond the plaquette), topological charge density $q(x)$ (per-site, parity-odd, the interpretability substrate). `train_gelt.py` currently trains on $q(x)$.]

6.4 Results: GELT vs. L-CNN vs. CNN

[TODO: THE MATCHED-PARAMETER SHOOTOUT – still to be run systematically. Planned figure: R^2 (or standardized MSE) per target per

model, plus learning curves. Include the non-equivariant CNN as the lower anchor and check the L-CNN reproduces Favoni-like behaviour. Also report the `mode="average"` vs. `mode="single"` transport A/B here (does path averaging dilute a specific-path target like an off-axis Wilson loop?).]

6.5 Gauge-invariance in practice

[TODO: Numerical invariance drift of trained models under random and worst-case Ω (float32 training vs. float64 checks) – the empirical companion to section 5.5.]

Chapter 7

Glueball spectroscopy with a learned variational operator

7.1 The idea: a network as a variational operator

[TODO: Frame shift from Ch. 6: instead of regressing a known observable, train GELT to BE an operator. `GELT(reduction="none")` emits a per-site scalar field $\rightarrow$ zero-momentum projection $\rightarrow$ timeslice operator $\bar{O}(t)$. The Rayleigh-quotient loss $-C(1)/C(0)$ is maximised exactly when the operator's ground-state overlap A_0 is maximised, and at the optimum the converged loss IS e^{-ma_t} : the loss value is a mass measurement. This is Morningstar–Peardon's variational principle [morningstar1999] with a neural network replacing the operator basis.]

7.2 The classical baseline

7.2.1 Toolkit

[TODO: Spatial-only APE smearing (gauge covariant, unit-tested; time links untouched so the transfer-matrix reading holds), the 0^{++} operator (sum of spatial-plane Wilson loops), vacuum-subtracted correlator, multi-level GEVP over cumulative smearing levels (eigh-whitening with an eigenvalue floor – low statistics can make $C(t_0)$ indefinite), jackknife everywhere.]

7.2.2 Run-by-run: the isotropic dead end and the anisotropic fix

[TODO: Tell it honestly as a negative-then-positive result. Isotropic $L = 12$, $\beta = 2.4$, $N = 2000$: no plateau – weak $ma \approx 0.8$ drowns by $\Delta \approx 3$ even with the GEVP; the LATTICE, not the operator basis, was the bottleneck. Anisotropic $L = 12$, $L_t = 24$, $\beta = 2.4$, $\xi = 3.0$, $N = 2000$: clean decay over 10–12 slices, GEVP ground state plateaus at $m_{\text{eff}}(\Delta=1) = 0.365 \pm 0.008$, $m_{\text{eff}}(\Delta=2) = 0.333 \pm 0.011$, i.e. $ma_t \approx 0.33$ – THE ANCHOR every learned result is judged against. Caveat box: $ma_s = \xi ma_t \approx 1.0$ at $\beta_s = 0.8$

is strong-coupling, not continuum physics; no ξ tuning, no continuum extrapolation (section 9.3).]

7.3 Constraints on the learned operator

[TODO: The theory-driven design decisions (from the 2026-07-01 audit) – worth a full section because they are where the physics disciplines the ML: (i) the operator must be **PER-TIMESLICE 3D** (**GELT(D=3)** on each timeslice’s spatial links, batch = config $\times$ timeslice) – any temporal receptive field voids the variational bound and makes the Rayleigh loss gameable toward $m \rightarrow 0$ ("never smear in time" applied to the network itself); (ii) hard train/held-out config split, blocked jackknife on held-out configs only; (iii) **mlp_zero_init** must be **OFF** (zero init $\Rightarrow$ identically zero Rayleigh gradient).]

7.3.1 Making the loss trainable

[TODO: The two loss pathologies and their fixes (real lessons, keep them): chained-ratio denominators replaced by $C(0)$; and the $\bar{O} \rightarrow \lambda \bar{O}$ flat direction, which site-coherent smeared inputs pump to float32 overflow, pinned by a $(\log C(0))^2$ scale regulariser.]

7.4 Training runs

7.4.1 Run 4 — thin input: a genuine operator that loses

[TODO: Thin-plaquette input: **GELT** learns $\approx$ **APE $\times 2$** ’s staple content (a depth-4 loop degree cannot rebuild iterated smearing) and is variationally redundant with the classical basis. Valuable as a capacity-vs-physics lesson.]

7.4.2 Run 5 — smeared input channels: the physics fix

[TODO: Multi-level smeared input channels (**INPUT_SMEAR_LEVELS** = (0, 2, 4, 6), $3 \times n_{\text{levels}}$ input channels). Results: Rayleigh loss saturates the transfer-matrix bound at the anchor (val $-0.6185 \leftrightarrow m a_t \approx 0.33$); m_{eff} plateaus from $\Delta = 1$; beats the classical **GEVP** at $\Delta = 1$ by 3.9σ (correlated blocked jackknife of the difference on shared test configs: -0.028 ± 0.007); the enlarged 5-operator **GEVP** (classical basis + **GELT**) collapses onto pure **GELT** – the learned operator already spans the useful directions.]

7.5 Quoting the result: cosh fits and the overlap A_0

[TODO: The **LGT-standard presentation**: masses from a correlated cosh fit (profiled- A grid fit, refit inside every jackknife sample), operator quality as the ground-state overlap fraction $A_0 = A(1 + e^{-mN_t})/C(0)$ – exactly the quantity the Rayleigh loss maximises. Comparator: the **PROJECTED**

classical GEVP operator (fixed ground-state vector v_0), the optimal single operator in the classical basis's span – the apples-to-apples opponent. Numbers (window $\Delta \in [2, 7]$, 400 held-out test configs):]

Operator	$m a_t$	A_0
GELT	0.332 ± 0.027	0.903 ± 0.047
GEVP-projected	0.340 ± 0.030	0.837 ± 0.056
correlated $\Delta A_0 = +0.066 \pm 0.031$ (2.1σ), Δm consistent with 0		

[TODO: Interpretation sentence: same mass, more ground-state weight – the network is a better operator, not different physics. Figure: glueball_overlap.png (m_{eff} + fit bands; $\rho(\Delta)$ overlap panel, flat at $A_0 \Leftrightarrow$ pure ground state).]

7.6 Replication on an independent ensemble

[TODO: Fresh ensemble (new seed, sampled from scratch) + from-scratch training. Results: GELT $A_0 = 1.013 \pm 0.062$ vs. GEVP-projected 0.925 ± 0.071 ; correlated $\Delta A_0 = +0.089 \pm 0.030$ (2.9σ); m_{eff} difference at $\Delta = 1$: -0.038 ± 0.008 (4.5σ). Combined with Run 5, the headline: $\Delta A_0 = +0.078 \pm 0.022$ (3.6σ). Key methodological lesson to write up: the Rayleigh floor is ENSEMBLE-SPECIFIC – the replication ensemble reads $\sim 1\sigma$ heavier (GEVP flat at ≈ 0.39 ; best val $-0.5638 \Leftrightarrow m \approx 0.395$, saturated to ~ 0.004 like Run 5's $-0.6185 \Leftrightarrow 0.33$), so LOSS VALUES do not replicate but OPERATOR QUALITY does. [TODO: if the seed-2 ensemble is rerun (the first attempt died on the Creutz rejection cap, since fixed), add it as a third independent measurement – a bonus, not a blocker.]]

7.7 Open items in this chapter

[TODO: Declared before the referee asks: (i) the matched-parameter L-CNN baseline on the SAME per-timeslice variational task – the last missing column of the shootout; (ii) anisotropy tuning / continuum limit; (iii) a third ensemble.]

Chapter 8

Interpreting attention as a physical measurement

8.1 Why equivariant attention is interpretable

[TODO: The chapter thesis: because the attention score is gauge INVARIANT by construction, the attention map $\alpha(x, \Delta x)$ is a legitimate physical observable – a measurement the network performs on the configuration, not an artifact of a gauge choice. Contrast with interpreting raw weights of a non-equivariant network (gauge-variant, meaningless). Define the extraction protocol: `_last_alpha` per layer/head under `no_grad`.]

8.2 Study 1: emergent correlation length

[TODO: PLANNED. Define an attention range ℓ_{att} (e.g. the decay length or first moment of α vs. $|\Delta x|$), measure it across β , and compare with the physical correlation length $\xi(\beta)$ from lattice ground truth. Strong version needs β -conditioned training $\Rightarrow \beta$ must enter the dataset (blocker recorded in section 9.3). Deliverable figure: $\ell_{\text{att}}(\beta)$ vs. $\xi(\beta)$.]

8.3 Study 2: localization on topological structure

[TODO: PLANNED. For a network trained on $q(x)$: overlay attention mass on the cooled/smear topological charge density; quantify with a correlation coefficient between per-site attention received and $|q(x)|$ after cooling. Blocker: cooling/smearing of $q(x)$ is not implemented yet. Include an intervention test (ablate heads / zero attention on lump sites and measure the regression degradation) so the claim is causal, not correlational.]

8.4 Study 3: head and layer specialization

[TODO: PLANNED. Per-head statistics of α (range, anisotropy, entropy) across layers; head-ablation matrix; do RoPE and bias variants specialise differently? Deliverable: a specialization taxonomy figure.]

8.5 Validation methodology

[TODO: The discipline that separates this from ML-paper attention-gazing: every claim validated against a lattice ground truth ($\xi(\beta)$ from correlators, cooled $q(x)$) and backed by an intervention. State the null results policy: negative findings get reported.]

8.6 Compute constraints

[TODO: The memory gate: physical receptive fields need larger R , which the current materialised L^1 -ball transport cannot afford – offset-chunked attention + on-the-fly transport is the enabling engineering item (section 9.3).]

Chapter 9

Conclusions and outlook

9.1 Summary of results

[TODO: One paragraph per delivered claim, mirroring section 1.3: (i) an exactly gauge-equivariant attention architecture with configuration-dependent receptive fields; (ii) validated samplers and a classical spectroscopy pipeline reproducing exact results and a clean GEVP anchor $m a_t \approx 0.33$; (iii) the learned variational operator result – same mass as the classical GEVP ($m a_t = 0.332 \pm 0.027$), better ground-state overlap ($\Delta A_0 = +0.078 \pm 0.022$, 3.6σ , two independent ensembles), with the enlarged GEVP collapsing onto the learned operator; (iv) [TODO: interpretability findings once they exist].]

9.2 Limitations

[TODO: Say them before the committee does: bare (untuned) anisotropy, so $m a_s \approx 1.0$ at $\beta_s = 0.8$ is not continuum physics; SU(2) only, single (β, ξ, L) point; the L-CNN column of the variational shootout is missing; the trained (RoPE) variant has weaker test coverage than the bias variant; no worst-case- Ω stress test; regression shootout (Ch. 6) not yet systematic.]

9.3 Future work

[TODO: Ordered roadmap (tracks notes/fable_audit.md; update as items land):]

1. **Architecture hygiene:** merge `blocks_bias` / `blocks_rope` into one module with a `pos_encoding` switch; parametrize equivariance tests over all variants; enforce RoPE coverage $d_{qkv} \geq 2D$; worst-case- Ω stress test.
2. **Scaling engineering:** offset-chunked attention + on-the-fly transport – the memory gate on the whole interpretability program at physical R .
3. **Complete the shootouts:** matched-parameter L-CNN on the per-timeslice variational glueball task; systematic regression comparison (Ch. 6).

4. **Interpretability program** (Ch. 8): β in the datasets, cooling/smearing of $q(x)$, then the three studies.
5. **Toward QCD:** U(1) and SU(3) samplers (Cabibbo–Marinari heat bath), 3+1D production, anisotropy tuning (renormalized ξ), continuum extrapolation of the 0^{++} mass; further channels (2^{++} , 0^{-+}) via irrep-projected learned operators.
6. **[TODO: Longer horizon, if space permits: learned operators for multi-level algorithms, transfer across β /volumes, generative directions.]**

9.4 Closing perspective

[TODO: One page: the broader claim – exact symmetry + adaptive attention turns a neural network from a black-box regressor into a variational instrument whose internal state (the attention map) is itself a physical measurement. Where this sits in the ML-for-LGT landscape.]

Appendix A

Notation and lattice conventions

A.1 Notation

[**TODO: Symbol table:** $U_\mu(x)$, $P_{\mu\nu}(x)$, $\Omega(x)$, $T_{\Delta x}(x)$, $\bar{O}(t)$, $C(\Delta)$, m_{eff} , A_0 , β , ξ , a_t , a_s , R (transport radius), D , L , L_t , N_c .]

A.2 Lattice conventions

[**TODO: The normative list (mirrors the code):** plaquette $P_{\mu\nu}(x) = U_\mu(x)U_\nu(x + \hat{\mu})U_\mu^\dagger(x + \hat{\nu})U_\nu^\dagger(x)$ with $\mu < \nu$ lexicographic ordering; Wilson action $S = \beta \sum_p (1 - \text{Re Tr } P/N_c)$; time is lattice axis 0, spatial axes $1 \dots D-1$; periodic boundary conditions; tensor layouts – links $(D, L, \dots, L, n_c, n_c)$, plaquettes $(D(D-1)/2, L, \dots, L, n_c, n_c)$; color axes always present, including $n_c = 1$ for $\mathbb{Z}_2$.]

Appendix B

Equivariance proofs

B.1 Gauge covariance of the shortest-path-averaged transport

[TODO: Induction over $|\Delta x|_1$ in the DP recursion: each extension step multiplies by $U_\mu(x)$ (forward) or $U_\mu^\dagger(x - \hat{e}_\mu)$ (backward), both of which transform covariantly, and covariance survives the average. Also prove the octant identity $T_{-\Delta x}(x) = T_{\Delta x}^\dagger(x - \Delta x)$. Tests: `tests/test_transport.py`.]

B.2 Equivariance of the GEMHSA layer

[TODO: Layer-by-layer: channel projections commute with the adjoint action (no color mixing); transported keys/values transform in the adjoint at the query site; the score $\text{Re Tr}[Q^\dagger \tilde{K}]$ is invariant (cyclic trace + unitarity), and positional encodings act only on the invariant score; the bilinear value $Q_v^\dagger \tilde{V}$ transforms in the adjoint; trace gate and residual preserve it. Tests: `tests/test_blocks.py` (SU(2) complex128, $\mathbb{Z}_2$ float64 bit-exact).]

B.3 Invariance of the readout and the loss

[TODO: Trace $\rightarrow$ MLP $\rightarrow$ reduction is invariant given adjoint inputs; hence $\bar{O}(t)$, $C(\Delta)$, and the Rayleigh loss are gauge invariant. Corollary worth stating: the attention map $\alpha(x, \Delta x)$ is gauge invariant – the license for Ch. 8.]

Appendix C

Codebase and reproducibility

C.1 The `gelt` package

[TODO: Module map (one paragraph each): `lattice.py` (groups, plaquettes, action, topology, transport), `sampler.py` (Metropolis, heat-bath + OR, anisotropy, τ_{int}), `glueball.py` (APE, correlators, GEVP, cosh fits, A_0), `blocks_rope.py` / `blocks_bias.py` (GELT), `lcnn.py` (L-CNN baseline), `cnn_baseline.py`, `data.py`. Entry-point scripts and the pytest suite; which test guards which thesis claim (cross-reference Appendix B).]

C.2 Production runs and hyperparameters

[TODO: One table per production result: ensemble parameters ($L = 12$, $L_t = 24$, $\beta = 2.4$, $\xi = 3.0$, $N = 2000$; n_{skip} from the τ_{int} pre-flight; `sampler = heat-bath +  $n_{\text{or}}$  OR`), GELT hyperparameters for Run 5 (smear levels (0, 2, 4, 6), depth, heads, d_{qkv} , optimizer, epochs), seeds and the RUN_TAG artifact-naming scheme, fit window $\Delta \in [2, 7]$, test split size (400 configs). Hardware: V100 for sampling/training, CPU for the offline fits.]

C.3 Reproduction recipe

[TODO: The command sequence: `check_glueball_autocorrelation.py` $\rightarrow$ `measure_glueball.py` (classical anchor, caches the ensemble) $\rightarrow$ `train_glueball.py` (Rayleigh training; env overrides GLUEBALL_ENSEMBLE_SEED / GLUEBALL_INIT_SEED / GLUEBALL_EVAL_ONLY) $\rightarrow$ `fit_glueball_overlap.py` (cosh fits + A_0 from the test-split $\bar{O}$ dump). The replication batch: `overnight_replication.sh`.]

